BreatheEasy

Tomorrow's air forecast next to your asthma action plan.

DSH Hacks V2 · AI × Healthcare
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4.8 million
U.S. children (6.7%) have asthma; 47% had an attack in the past year (CDC, 2023 NHIS)

6th worst
Houston-Pasadena metro for ozone in the U.S. (American Lung Association, State of the Air 2026)

43.3 days/yr
weighted unhealthy ozone days, Harris County, 2022–2024 (ALA)

The problem

Doctors give families a paper **asthma action plan** (green, yellow, red zones). On bad-ozone days EPA tells people with asthma to "follow their action plan and keep quick-relief medicine handy". But the plan sits in a drawer and the forecast lives in another app, so they rarely meet the night before, while there's still time to move practice indoors or take the pre-exercise medicine already prescribed.

What BreatheEasy does

- **Tomorrow's AQI, the EPA way.** Hourly ozone and PM2.5 forecasts (Open-Meteo / CAMS) turned into a daily U.S. AQI with EPA's breakpoints and averaging rules (8-h ozone, 24-h PM2.5, 2024 update).
- **Guidance in your doctor's words.** EPA's advice for that level beside the text you copied from your own plan. It never invents medicines or doses.
- **20-second check-in.** Peak flow and symptoms → your zone (NHLBI: 80% / 50% of personal best), that zone's steps, and Call 911 on danger signs.
- **On-device model that explains each prediction.** Logistic regression trains in the browser on your own check-ins (ozone, PM2.5, pollen, heat, humidity, yesterday) and explains each prediction feature by feature. Silent until 14 days of data.
- **Any city.** Type a city (Open-Meteo Geocoding): 5 matches, biggest first, remembered next visit. Not just Houston.
- **Demo button.** A button on the Air card fills the personal outlook with demo data tagged **SYNTHETIC**, so no 14-day wait to try it.
- **Explain this day.** Pick any logged day: each condition's push (odds multiplier) and what a model trained only on earlier days would have said. What it weighed, not what caused the day.
- **Share with coach or nurse.** One tap writes an editable note (forecast, EPA steps, your pre-exercise line, 911 line); plus calendar and print.
- **Offline, private, English + Spanish.** Installable; with no signal it shows only a forecast saved in the last 3 hours, marked with its age. No account, server or tracking; one-tap export and delete. Spanish uses CDC, NHLBI and AirNow terms.



Tomorrow card on a real forecast (Sugar Land, TX, Oct 10, 2026): EPA-method AQI 140; Open-Meteo's own peak hourly AQI was 143. The personal outlook here uses the labelled synthetic demo data.

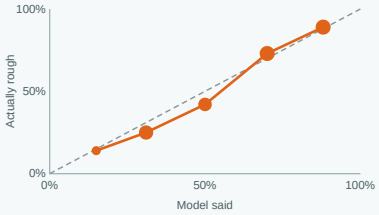
Limits and next steps

Forecasts are model estimates, not monitor readings. No free U.S. pollen forecast exists, so users log the Houston Health Department count (labelled as logged). Next: pilot with families and a school nurse, AirNow Ozone Action Day alerts, and reminders.

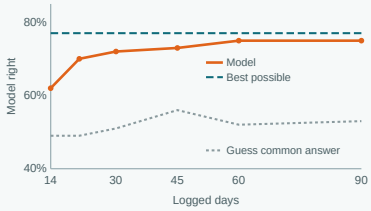
How much can you trust it?

SYNTHETIC check: made-up people, not patients

Calibration after 60 logged days



Accuracy vs. days logged



Built into the app and run in the browser: 12 made-up people with a hidden pattern, 200 days each; the model trains on logged days and is scored only on the 100 days after.

At 60 days (1,200 held-out days): **calibration error 3.8 pts** (said 57% rough on average; 55% were). **Brier 0.172** vs. 0.252 for each person's usual rate and 0.157 best possible. **75% right** vs. 52% guessing the common answer (77% best).

Runs slightly high in the middle bins. Real people are messier; this checks the method, not how it does on real patients.

How it's built and tested

Static HTML/CSS/JavaScript: no framework, no keys, runs on any phone; a content-hashed service worker for offline use. **125 unit tests** (incl. EPA's worked ozone example and every breakpoint edge) and **142 headless-Chromium checks** (zones, 911 alert, model, explain view, trust card, share note, city search, one-tap demo, Spanish, offline reload, 390 px layout, API failure).

Not medical advice; not a medical device. BreatheEasy helps you follow the plan your doctor wrote. Trouble walking or talking from shortness of breath, or lips or fingernails that are blue, pale, or gray: call 911 (CDC, 2024). *En español:* problemas para caminar o hablar por la dificultad para respirar, o labios o uñas azules, pálidos o grises: llama al 911.

Sources: CDC Most Recent National Asthma Data (2023 NHIS) · American Lung Association, State of the Air 2026 (Texas) · NHLBI Asthma Action Plan (2020) · CDC Plan de acción para el control del asma (2024) · U.S. EPA, 40 CFR Part 58 App. G and Air Quality Guide for Ozone · Open-Meteo Air Quality API (CAMS, CC BY 4.0). Built with help from an AI coding assistant; every statistic checked against its source.